

Diffusion-Based Spatio-Temporal Channel Prediction via Non-Stationarity Decoupling

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Abstract—Accurate channel prediction is critical for mitigating the impact of channel aging in high-mobility environments. One of the primary challenges in channel prediction lies in the rapid variations of wireless channels, which lead to significant temporal fluctuations in the data distribution and result in pronounced non-stationary characteristics. Existing methods are limited in their ability to explicitly model or address the non-stationary nature of time-varying channels. To overcome this limitation, we introduce the Diffusion-based Spatio-Temporal Channel Prediction method via Non-stationarity Decoupling (DST-ND). Specifically, the DST-ND approach incorporates a non-stationarity decoupling module based on moving average. This module effectively separates time-varying channels into their non-stationary and stationary components. Additionally, a spatio-temporal diffusion model is employed to capture both temporal and spatial dependencies in channels, utilizing the global receptive field enabled by the attention mechanism. Experimental results on both real-world measurement data and simulation data demonstrate that the DST-ND method achieves state-of-the-art prediction performance across a variety of high-mobility scenarios.

Index Terms—Channel prediction, wireless communication, non-stationarity decoupling, spatio-temporal correlations, diffusion models.

I. INTRODUCTION

MILLIMETER-WAVE (mmWave) massive multiple-input multiple-output (MIMO) is a key technology for fifth-generation (5G) and beyond wireless communication systems [1]. In such systems, a major challenge is channel aging, which refers to the time-varying nature of wireless channels due to user mobility and environmental dynamics [2], [3].

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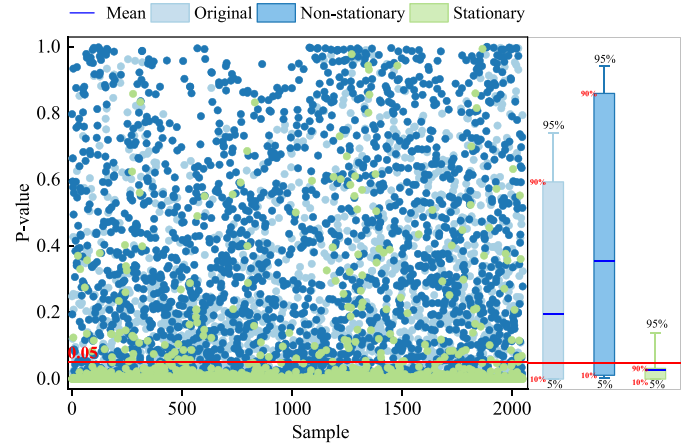


Fig. 1. P-value results of the ADF test on the NLOS30 dataset, which contains 2,048 samples. **Stationarity Assumption:** a p-value less than 0.05 indicates that the sequence is stationary. Detailed results are presented in table V.

This effect is particularly severe in high-mobility scenarios, where rapid changes degrade the accuracy of beamforming and resource allocation, ultimately compromising system performance [4]. Therefore, accurate channel state information (CSI) prediction, hereafter referred to as channel prediction, is critical to mitigate the impact of channel aging and maintain link reliability [5].

Existing channel prediction methods can be broadly categorized into traditional parametric models [6], [7], [8], [9], [10], [11] and data-driven deep learning approaches [12], [13], [14], [15], [16], [17]. However, predicting real-world time-varying channels remains challenging due to two intertwined issues: temporal non-stationarity and complex spatial correlation. Parametric methods often rely on idealized stationary assumptions, failing to adapt to the pronounced non-stationary characteristics of channels in dynamic environments [18]. Deep learning methods, such as Recurrent Neural Networks (RNNs) [19] and Long Short-Term Memory (LSTM) networks [20], can learn from data but are prone to error accumulation in rolling prediction paradigms [11]. While recent Transformer-based parallel prediction frameworks alleviate error accumulation [4], [16], [17], they still lack explicit mechanisms to handle the inherent non-stationarity and its complex interplay with spatial dependencies across antenna arrays. As shown in Fig. 1, over 56% of real-world CSI sequences exhibit significant non-stationary behavior, underscoring the pervasiveness of this challenge. Crucially,

these two challenges, temporal non-stationarity and spatial correlation, are not isolated. Rapid variations in channel statistics not only cause unpredictable temporal dynamics but also induce dynamic shifts in spatial structures. Consequently, existing methods often inadequately capture their interaction, leading to incomplete modeling and reduced prediction accuracy in practical, high-mobility scenarios [21], [22], [23], [24], [25], [26].

To address these challenges, this paper proposes Diffusion-based Spatio-Temporal Channel Prediction method via Non-stationarity Decoupling (**DST-ND**). The core idea is to explicitly decouple and separately model the non-stationary and stationary components within the CSI. First, a Non-Stationarity Decoupling (NS-Dec) module, utilizing a predefined Moving Average Kernel (MAK), separates the CSI into two distinct parts. This decomposition, validated in Fig. 1, effectively isolates non-stationary dynamics, with the stationary component showing significantly reduced non-stationary behavior. For the non-stationary component, a simple parallel-prediction network is employed. For the stationary component containing the main CSI, we propose a Spatio-Temporal Diffusion Model (STDM). Built upon a diffusion probabilistic framework [27], STDM learns to forecast the probability distribution of future CSI conditioned on past observations, naturally providing uncertainty predictions [28]. Furthermore, STDM incorporates Transformer layers [29] within its denoising network to jointly capture the complex temporal and spatial dependencies.

To validate the effectiveness of the proposed method, we conducted extensive experiments on a series of high-dynamic channel datasets. NS-Dec module proves to be effective. As illustrated in Fig. 1, after decoupling, the majority of non-stationary characteristics are successfully isolated into the non-stationary component. Furthermore, the experimental results demonstrate that compared to state-of-the-art (SOTA) methods, such as the Transformer-based parallel prediction method [4], [16] and the RNN-based rolling prediction method [11], DST-ND achieves significant improvements in prediction accuracy. Specifically, compared to deep learning methods, DST-ND improves prediction performance by 13.6% to 36.7% in various high-dynamic scenarios. It is worth noting that our method also provides probabilistic predictions based on diffusion models, which are absent in existing channel prediction methods.

The main contributions of this paper are summarized as follows:

- We propose DST-ND, a novel diffusion-based spatio-temporal channel prediction method that leverages non-stationarity decoupling, achieving SOTA prediction performance across diverse high-mobility scenarios.
- Specifically, we design NS-Dec to separate the channel into non-stationary and stationary components using a moving average kernel, effectively addressing abrupt distribution shifts.
- We further develop STDM, which leverages a diffusion model to learn the conditional distribution of CSI, providing both predictive accuracy and uncertainty estimation.

TABLE I
NOTATION SUMMARY

Notation	Definition
<i>System & Channel Parameters</i>	
$\mathbf{H}^t \in \mathbb{C}^{N_t \times N_r}$	Channel matrix at time t with N_t transmit and N_r receive antennas
$\mathcal{H} \in \mathbb{C}^{L \times N_t \times N_r}$	Historical channel tensor over lookback window length L
L, T, K	Lookback window, prediction horizon, feature dimension ($K = 2N_tN_r$)
$\mathbf{X} \in \mathbb{R}^{K \times L}$	Vectorized input channel matrix
$\hat{\mathcal{H}} \in \mathbb{C}^{T \times N_t \times N_r}$	Predicted channel tensor
<i>Decomposition</i>	
$\mathbf{X}^{\text{nst}}$	Non-stationary component
$\mathbf{X}^{\text{st}}$	Stationary component
$\hat{\mathbf{Y}}^{\text{nst}}, \hat{\mathbf{Y}}^{\text{st}}$	Predicted non-stationary and stationary components
<i>Diffusion Model</i>	
N	Total diffusion steps
$\mathbf{x}_n$	Noisy sample at step n
ϵ_θ	Noise prediction network
α_n, β_n	Variance schedule parameters
$\bar{\alpha}_n = \prod_{i=1}^n \alpha_i$	Cumulative variance product
<i>Model Components</i>	
NS-Dec	Non-Stationarity Decoupling module
STDM	Spatio-Temporal Diffusion Model
T-extractor	Temporal correlation extractor
S-extractor	Spatial correlation extractor

- To enhance spatio-temporal modeling, we incorporate a novel denoising network with two Transformer layers in STDM, achieving an average Normalized Mean Square Error (NMSE) improvement of 26.9% compared to temporal-only baselines.

Notation: For clarity, we use bold uppercase letter to denote matrix, bold lowercase letter to denote vector, and non-bold letter to denote scalar. $\mathbf{x}^{(t_1:t_2)}$ denotes a continuous sequence from time t_1 to t_2 . $(\cdot)^\top$ is the transpose of a matrix. $\mathbf{X}[i, j]$ denotes the element at i -th row and the j -th column of $\mathbf{X}$. The complex number field is represented by $\mathbb{C}$ and the real number field is represented by $\mathbb{R}$. $\text{vec}(\cdot)$, $\text{dvec}(\cdot)$ denote the vectorization of a matrix and the inverse process, respectively. $\Re(\cdot)$, $\Im(\cdot)$ denotes the real and imaginary parts of a matrix, respectively. The symbol Θ denotes the set of model parameters and $\mathbb{E}\{\cdot\}$ denotes the statistical expectation. $[\mathbf{x}, \mathbf{y}]$ denotes the concatenation of $\mathbf{x}$ and $\mathbf{y}$. More notations are summarized in Table I.

Organization: The remainder of this paper is organized as follows. Section II introduces the preliminaries of our method, including the denoising diffusion probabilistic models and the linear attention transformer. Section III presents the system model and formulates the channel prediction problem under non-stationary conditions. In Section IV, we detail the proposed DST-ND method, including the NS-Dec and the STDM, along with the corresponding loss functions. Section V elaborates on the implementation of STDM and its spatio-temporal awareness modeling. Simulation results and performance comparisons with baseline methods are provided in Section VI, followed by ablation studies and visualization analyses. Conclusions are drawn in Section VII. Moreover, the Appendix provides additional implementation details.

II. PRELIMINARIES

In this section, we review the Denoising Diffusion Probabilistic Models (DDPMs) and the Linear Attention Transformer (LAT) architecture. These two components serve as the core blocks of our proposed method, aiming at capturing probabilistic dynamics and spatio-temporal dependencies for time-varying wireless channels.

A. Denoising Diffusion Probabilistic Models

Let $\mathbf{x}_0 \sim q_{\mathcal{X}}(\mathbf{x}_0)$ denote the training vector form input space $\mathcal{X} = \mathbb{R}^D$, and let $p_{\theta}(\mathbf{x}_0)$ denote the probability density function (PDF) which aims to approximate $q_{\mathcal{X}}(\mathbf{x}_0)$. DDPMs [27] are latent variable models that learn to generate samples from a data distribution by modeling the reverse diffusion process. Specifically, DDPMs define a forward diffusion process that gradually adds Gaussian noise to the data $\mathbf{x}_0, \dots, \mathbf{x}_N$ via a Markov chain [30], with each variable in $\mathbb{R}^D$ as follows

$$\begin{aligned} q(\mathbf{x}_n|\mathbf{x}_{n-1}) &:= \mathcal{N}(\mathbf{x}_n; \sqrt{1 - \beta_n}\mathbf{x}_{n-1}, \beta_n\mathbf{I}), \\ q(\mathbf{x}_{1:N}|\mathbf{x}_0) &:= \prod_{n=1}^N q(\mathbf{x}_n|\mathbf{x}_{n-1}), \end{aligned} \quad (1)$$

where $\beta_n \in (0, 1)$ is a variance schedule that controls the amount of noise added at each diffusion step n , and $\mathbf{I}$ denotes the identity covariance matrix, where all elements on the diagonal are 1 and all other elements are 0. The joint $p_{\theta}(\mathbf{x}_{0:N})$ is called the reverse process, and is defined as a Markov chain with learned Gaussian transitions starting with $p(\mathbf{x}_N) = \mathcal{N}(\mathbf{x}_N; \mathbf{0}, \mathbf{I})$, where each subsequent transition of

$$p_{\theta}(\mathbf{x}_{n-1}|\mathbf{x}_n) := \mathcal{N}(\mathbf{x}_{n-1}; \mu_{\theta}(\mathbf{x}_n, n), \sigma_{\theta}(\mathbf{x}_n, n)) \quad (2)$$

is given by a parametrization of our choosing denoted by

$$p_{\theta}(\mathbf{x}_{n-1}|\mathbf{x}_n) := \mathcal{N}(\mathbf{x}_{n-1}; \mu_{\theta}(\mathbf{x}_n, n), \sigma_{\theta}(\mathbf{x}_n, n)\mathbf{I}), \quad (3)$$

with shared parameters θ . As shown in [27], a property of the forward process is that it admits sampling $\mathbf{x}_n$ at any arbitrary noise level n in closed form, since if $\alpha_n := 1 - \beta_n$ and $\bar{\alpha}_n := \prod_{i=1}^n \alpha_i$ its cumulative product, we have

$$q(\mathbf{x}_n|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_n; \sqrt{\bar{\alpha}_n}\mathbf{x}_0, (1 - \bar{\alpha}_n)\mathbf{I}). \quad (4)$$

Thus, using the reparameterization trick in [31] and defining $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, we have

$$\mathbf{x}_n = \sqrt{\bar{\alpha}_n}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_n}\epsilon. \quad (5)$$

The goal of p_{θ} is to eliminate the noise added in the forward process. The θ are learned by minimizing the variational lower bound (VLB) of the log-likelihood of the data, which can be expressed as

$$\begin{aligned} \mathcal{L}_{\text{vlb}} : &= \underbrace{-\log p_{\theta}(\mathbf{x}_0|\mathbf{x}_1)}_{\mathcal{L}_0} \\ &+ \sum_{n=2}^N \underbrace{D_{\text{KL}}(q(\mathbf{x}_{n-1}|\mathbf{x}_n, \mathbf{x}_0) \parallel p_{\theta}(\mathbf{x}_{n-1}|\mathbf{x}_n))}_{\mathcal{L}_{n-1}} \\ &+ \underbrace{D_{\text{KL}}(q(\mathbf{x}_N|\mathbf{x}_0) \parallel p(\mathbf{x}_N))}_{\mathcal{L}_N}. \end{aligned} \quad (6)$$

[27] showed that by the (4), the forward process posterior in theses Kullback-Leibler (KL) divergences [32] when conditional on $\mathbf{x}_0$, i.e., $q(\mathbf{x}_{n-1}|\mathbf{x}_n, \mathbf{x}_0)$ are tractable given by

$$q(\mathbf{x}_{n-1}|\mathbf{x}_n, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{n-1}; \frac{\sqrt{\bar{\alpha}_{n-1}}\beta_n}{1 - \bar{\alpha}_n}\mathbf{x}_0 + \frac{\sqrt{\bar{\alpha}_n}(1 - \bar{\alpha}_{n-1})}{1 - \bar{\alpha}_n}\mathbf{x}_n, \frac{1 - \bar{\alpha}_{n-1}}{1 - \bar{\alpha}_n}\beta_n\mathbf{I}). \quad (7)$$

$$\begin{aligned} D_{\text{KL}}(q(\mathbf{x}_{n-1}|\mathbf{x}_n, \mathbf{x}_0) \parallel p_{\theta}(\mathbf{x}_{n-1}|\mathbf{x}_n)) \\ = \mathbb{E}_q \left[\frac{1}{2\sigma_{\theta}} \|\hat{\mu}_n(\mathbf{x}_n, \mathbf{x}_0) - \mu_{\theta}(\mathbf{x}_n, n)\|^2 \right] + C, \end{aligned} \quad (8)$$

where C is a constant which does not depend on θ . Finally, [27] found predicting ϵ worked best, especially when combined with a reweighted loss function:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{n, \mathbf{x}_0, \epsilon} \left[\|\epsilon - \epsilon_{\theta}(\mathbf{x}_n, n)\|^2 \right]. \quad (9)$$

They also acknowledge the possibility of directly predicting $\mathbf{x}_0$. It is important to note that estimating ϵ in (9) is equivalent to estimating a scaled version of the score function, which is the gradient of the log density of the noisy data

$$\begin{aligned} \nabla_{\mathbf{x}_n} \log q_n(\mathbf{x}_n|\mathbf{x}_0) &\approx -\frac{1}{1 - \bar{\alpha}_n}(\mathbf{x}_n - \sqrt{\bar{\alpha}_n}\mathbf{x}_0) \\ &= -\frac{1}{\sqrt{1 - \bar{\alpha}_n}}\epsilon. \end{aligned} \quad (10)$$

Thus, data generation via denoising relies on the score function and can be viewed as a form of noise-conditional score-based generation as described by [33]. Consequently, the reverse process $p_{\theta}(\mathbf{x}_{n-1}|\mathbf{x}_n)$ in the DDPMs (Denoising Diffusion Probabilistic Models) can be formulated as

$$\mathbf{x}_{n-1} = \frac{1}{\sqrt{\alpha_n}} \left(\mathbf{x}_n - \frac{1 - \alpha_n}{\sqrt{1 - \bar{\alpha}_n}} \epsilon_{\theta}(\mathbf{x}_n, n) \right) + \sigma_n z, \quad (11)$$

where σ_n denotes a hyperparameter, and z represents standard Gaussian noise.

B. Linear Attention Transformer

The linear attention Transformer (LAT), introduced in 2019, was designed to enhance the attention mechanism by linearizing its computation, thereby improving efficiency in processing long sequences [34]. The traditional attention mechanism [29] has a time complexity of $\mathcal{O}(n^2)$, and its computation is defined as

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax} \left(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\sqrt{d_k}} \right) \mathbf{V}, \quad (12)$$

where $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ denote the query, key, and value matrices, respectively, and d_k is the dimensionality of the key vectors. In contrast, LAT reduces the time complexity to $\mathcal{O}(n)$, which has significant implications for channel prediction. The LAT achieves this by avoiding the explicit computation of $\mathbf{Q}\mathbf{K}^{\top}$, instead leveraging a kernel function ϕ to project the Query and Key into a different space. The linear attention mechanism is then computed using

$$\text{LinearAttention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \phi(\mathbf{Q}) \cdot (\phi(\mathbf{K})^{\top} \cdot \mathbf{V}), \quad (13)$$

where $\phi(\mathbf{Q})$ and $\phi(\mathbf{K})$ are the projections of the query and key, respectively, obtained through the mapping function ϕ .

III. PROBLEM FORMULATION AND CURRENT CHALLENGES

In this section, we discuss the channel prediction problem in high-mobility measured Non-Line-of-Sight (NLOS) scenarios and the current challenges.

A. Problem Formulation of Channel Prediction

This paper investigates a wireless channel prediction task in high-mobility environments based on real-world channel measurements in datasets NLOS30 and NLOS60.¹ Specifically, these channel measurements are conducted in practical urban macrocell (UMa) scenarios under non-line-of-sight (NLOS) conditions, reflecting temporal evolution of wireless channels experienced by fast-moving users. We consider a single-user and multi-antenna (MIMO) cellular system with a single carrier operating in the FR1 band (2 GHz) [35]. According to [36], the system employs Frequency Division Duplexing (FDD) and Orthogonal Frequency Division Multiplexing (OFDM). The time is discretized into timeslots, indexing by $t = 1, 2, \dots$. We focus on the complex channel response matrix at time step t is denoted by:

$$\mathbf{H}^t = \begin{bmatrix} h_{1,1}^{(t)} & h_{1,2}^{(t)} & \cdots & h_{1,N_r}^{(t)} \\ h_{2,1}^{(t)} & h_{2,2}^{(t)} & \cdots & h_{2,N_r}^{(t)} \\ \vdots & \vdots & \ddots & \vdots \\ h_{N_t,1}^{(t)} & h_{N_t,2}^{(t)} & \cdots & h_{N_t,N_r}^{(t)} \end{bmatrix} \in \mathbb{C}^{N_t \times N_r} \quad (14)$$

where N_t denotes the number of transmitter antennas, N_r denotes the number of receiver antennas. Likewise, $h_{i,j}^{(t)} \in \mathbb{C}$ denotes the complex channel gain from the i -th transmitter antenna to the j -th receiver antenna at time t . To model the temporal evolution of channel coefficient matrix $\mathbf{H}^t$, we construct an input tensor over a historical window of length L :

$$\mathcal{H} = [\mathbf{H}^{t-L+1}, \dots, \mathbf{H}^t] \in \mathbb{C}^{L \times N_t \times N_r} \quad (15)$$

Given a specific input $\mathcal{H}$, we aim at predicting future CSI within the upcoming T , formulated as:

$$\hat{\mathcal{H}} = (\hat{\mathbf{H}}^{t+1}, \dots, \hat{\mathbf{H}}^{t+T}) = f_{\Theta}(\mathbf{H}^{t-L+1}, \dots, \mathbf{H}^t), \quad (16)$$

where $f_{\Theta}(\cdot)$ denotes the learning neural network model, Θ represents the model parameters.

Deep Learning Channel Prediction schemes often cannot receive complex numbers as inputs, so we vectorize the $\mathbf{H}^t$ matrix as $\mathbf{x}^t \in \mathbb{R}^K$ ($K = N_t \times N_r \times 2$), where t is the time index of the channel. We then vectorize them and concatenate them into a vector $\mathbf{h}^t$, as follows

$$\mathbf{x}^t = \left[\text{vec}(\Re[\mathbf{H}^t])^\top, \text{vec}(\Im[\mathbf{H}^t])^\top \right]^\top. \quad (17)$$

Then, we can know that $\mathbf{x}^t$, takes the form of

$$\mathbf{x}^t = (\mathbf{x}^t[1], \dots, \mathbf{x}^t[k], \dots, \mathbf{x}^t[K])^\top. \quad (18)$$

So the specific expression of channel prediction can be updated from

$$\hat{\mathbf{x}}^{t+1}, \dots, \hat{\mathbf{x}}^{t+T} = f_{\Theta}(\mathbf{x}^{t-L+1}, \dots, \mathbf{x}^t). \quad (19)$$

The core objective of channel prediction is to model the mapping function $f(\cdot)$. To construct f , extensive research has explored deep learning-based approaches, including rolling prediction and parallel prediction.

Rolling Prediction: Rolling prediction refers to autoregressive prediction, where the output is $\hat{\mathbf{x}}^{t+i} \in \mathbb{R}^K$. Using a window of past values to predict the next value, the window is then shifted, and the predicted value is used for the subsequent step. This process is repeated until T values are predicted. The window size is set to L , the length of the past channel, as

$$\hat{\mathbf{x}}^{t+i} = \sum_{l=1}^L \phi_l \mathbf{x}^{t+L-i-1} + b, \quad (20)$$

where i ranges from 1 to T , ϕ_l denotes the coefficient of the i -th term, and b is the bias term. Finally, all predicted values $\hat{\mathbf{x}}^{(t+1:t+T)}$ are obtained. The objective of rolling prediction is to determine the parameters $(\phi_1, \phi_2, \dots, \phi_L, b)$ that best fit the given set of T future channels.

Parallel Prediction: Parallel prediction refers to using multiple past channels to directly predict multiple future channels in a single step, rather than iteratively. Its output is $\hat{\mathbf{x}}^{t+1:t+T} \in \mathbb{R}^{K \times T}$. The parallel prediction process follows the form of (19). However, existing parallel prediction methods often exploit only temporal correlations, i.e., modeling each position of $\mathbf{x}$ independently, as

$$\hat{\mathbf{x}}[k]^{t+1}, \dots, \hat{\mathbf{x}}[k]^{t+T} = f_{\Theta}(\mathbf{x}[k]^{t-L+1}, \dots, \mathbf{x}[k]^t), \quad (21)$$

where k ranges from 1 to K .

B. Current Challenges

Definition 1 (Temporal Distribution Shift [37]): Given a time series $\mathbf{X} \in \mathbb{R}^{K \times L}$, by applying a sliding window, we obtain a set of time series segments of length M , denoted as

$$\mathcal{D} = \{X_{n,i:i+M} \mid n \in [1, N], i \in [1, L-M]\},$$

where each $X_{n,i:i+M}$ corresponds to a subsequence of $X_{n,\cdot}$. Temporal distribution shift refers to the case where $\mathcal{D}$ can be partitioned into S sets, i.e., $\mathcal{D} = \bigcup_{i=1}^S \mathcal{D}_i$, where each $\mathcal{D}_i$ denotes a subset with the same data distribution $P_{\mathcal{D}_i}(x)$, and for any $i \neq j$, $P_{\mathcal{D}_i}(x) \neq P_{\mathcal{D}_j}(x)$. In other words, the data distribution of the time series varies across different temporal segments, and such variation is referred to as **temporal distribution shift (TDS)**.

Current channel prediction methods face two fundamental challenges.

1) Challenge 1: Non-stationarity: Most existing methods assume channel stationarity, where statistical properties remain constant over time. However, in high-mobility scenarios channel characteristics evolve rapidly, inducing strong non-stationarity (or called TDS). This causes traditional methods to accumulate prediction errors. To quantify this issue, we applied the Augmented Dickey-Fuller (ADF) test to 2,048 real-world channel samples. As shown in Fig. 1, over 56% exhibited non-stationarity, providing empirical evidence that classical stationarity assumptions are violated. The key insight is that channels contain two components: non-stationary component and stationary component. Separate modeling of these components is essential for accurate prediction.

¹ www.mobileai-dataset.cn: CSI-02-0003

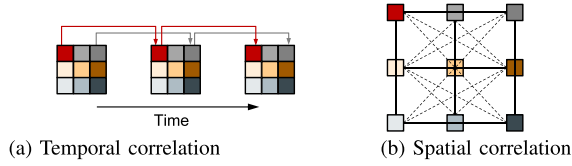


Fig. 2. Illustration of temporal and spatial correlations in channel state information (CSI). (a) **Temporal correlation** captures dependencies across consecutive time frames. (b) **Spatial correlation** reflects the interdependence among elements within a CSI frame, such as between antenna ports.

TABLE II

COMPARISON OF CHANNEL PREDICTION METHODOLOGIES: NEURAL NETWORK-BASED (NER.), PREDICTION STRATEGY, CORRELATION MODELING (TEMPORAL AND SPATIAL), NON-STATIONARITY DECOUPLING (NS-DEC), AND UNCERTAINTY FORECASTING

Model	Ner.	Pred Strategy	Temporal	Spatial	NS-Dec	Uncertainty
PVEC	×	Rolling	✓	×	×	×
FFT	×	Rolling	✓	×	×	×
Kalman	×	Rolling	✓	×	×	×
ARIMA	×	Rolling	✓	×	×	×
R. RNN	✓	Rolling	✓	×	×	×
R. LSTM	✓	Rolling	✓	×	×	×
P. Transformer	✓	Parallel	✓	×	×	×
LLM4CP	✓	Parallel	✓	×	×	×
STEM	✓	Parallel	✓	×	×	×
Ours	✓	Parallel	✓	✓	✓	✓

2) *Challenge 2: Incomplete Spatio-Temporal Correlation Modeling*: Most deep learning approaches exploit only temporal correlations while ignoring spatial dependencies across antennas or subcarriers (Fig. 2). CSI is inherently multi-dimensional; its values depend on both time and physical dimensions. In large-scale MIMO systems, spatial correlations are particularly strong. Ignoring these spatial structures prevents models from fully capturing channel dynamics, reducing prediction accuracy and robustness.

As summarized in Table II, our method uniquely addresses both challenges through: (1) non-stationarity decoupling to separate non-stationary component, (2) joint spatio-temporal correlation extraction via Transformer, and (3) probabilistic forecasting via diffusion models.

IV. DIFFUSION-BASED SPATIO-TEMPORAL CHANNEL PREDICTION METHOD VIA NON-STATIONARITY DECOUPLING

In this section, to alleviate the sensitivity to abrupt non-stationary variations in channel prediction, we propose DST-ND as shown in Fig. 3. The method consists of two interdependent modules: NS-Dec and STDM. First, NS-Dec employs a predefined MAK to extract the non-stationary component from the original time-varying channels and compute the stationary component. Then, the non-stationary component is processed via an LSTM with parallel prediction. Finally, STDM employs a diffusion model to generate the probability distribution of the stationary component.

A. Non-Stationarity Decoupling

To learn complex temporal patterns in the context of channel prediction, we adopt the concept of decomposition [38],

which can extract the non-stationary component from time-varying channels. These two parts respectively reflect the long-term changes and seasonality of the series. However, direct decomposition is not feasible for future frames, as the future is unknown. Specifically, we adjust the moving average to smooth out cyclical fluctuations and highlight the long-term trend. For input data $\mathbf{X} \in \mathbb{R}^{K \times L}$, the process is as follows

$$\begin{aligned} \mathbf{X}' &= \text{Padding}(\mathbf{X}), \\ \mathbf{X}^{\text{nst}} &= \text{AvgPool}(\mathbf{X}'), \\ \mathbf{X}^{\text{st}} &= \mathbf{X} - \mathbf{X}^{\text{nst}}, \end{aligned} \quad (22)$$

where $\mathbf{X}^{\text{nst}}, \mathbf{X}^{\text{st}}$ denote the non-stationary and stationary components respectively, $\text{Padding}(\cdot)$ pads both sides of the data so the original length is preserved after applying $\text{AvgPool}(\cdot)$, and $\text{AvgPool}(\cdot)$ refers to the moving average with a specified kernel size. A schematic illustration of the decoupling is presented in Fig. 4.

B. Non-Stationary Component Prediction by LSTM

The non-stationary component $\mathbf{X}^{\text{nst}} \in \mathbb{R}^{K \times L}$ is fed into a 2-layer LSTM network. For each time step τ in the sequence ($1 \leq \tau \leq L$), the LSTM updates its hidden state $\mathbf{s}_\tau \in \mathbb{R}^{d_h}$ and cell state $\mathbf{c}_\tau$ as:

$$\mathbf{s}_\tau, \mathbf{c}_\tau = \text{LSTM}(\mathbf{x}_\tau, \mathbf{s}_{\tau-1}, \mathbf{c}_{\tau-1}), \quad (23)$$

where d_h is the hidden dimension and $\mathbf{x}_\tau$ is the input feature vector at step τ . The LSTM outputs hidden states $\mathbf{S} = [\mathbf{s}_1, \dots, \mathbf{s}_L] \in \mathbb{R}^{d_h \times L}$. A linear projection layer then maps each hidden state to the prediction horizon:

$$\hat{\mathbf{Y}}^{\text{nst}} = \mathbf{W}_p \mathbf{S} + \mathbf{b}_p, \quad (24)$$

where $\mathbf{W}_p \in \mathbb{R}^{T \times d_h}$ and $\mathbf{b}_p \in \mathbb{R}^T$ are learnable weights and biases, producing predictions $\hat{\mathbf{Y}}^{\text{nst}} \in \mathbb{R}^{K \times T}$ for the T -step forecast horizon. The loss function for the non-stationary component is $\mathcal{L}_{\text{nst}} = \mathbb{E} [\|\mathbf{Y}^{\text{nst}} - \hat{\mathbf{Y}}^{\text{nst}}\|^2]$.

C. Stationary Component Prediction by Spatio-Temporal Diffusion Model

STDM is designed to predict the stationary component of the time-varying channel. Specifically, STDM employs a diffusion model to predict the probability distribution of time-varying channels, using historical CSI as the conditional input for the diffusion model. The working process of STDM can be simply expressed as follows

$$\hat{\mathbf{Y}}^{\text{st}} = \text{STDM}(\hat{\mathbf{X}}^{\text{st}}). \quad (25)$$

Then, its loss function (27) will be introduced in Section V, and here, we will simply denote it as $\mathcal{L}_{\text{st}}$ for the next subsection's discussion.

In order to provide a more detailed discussion on the working mechanism of STDM, we have elaborated on it in Section V.

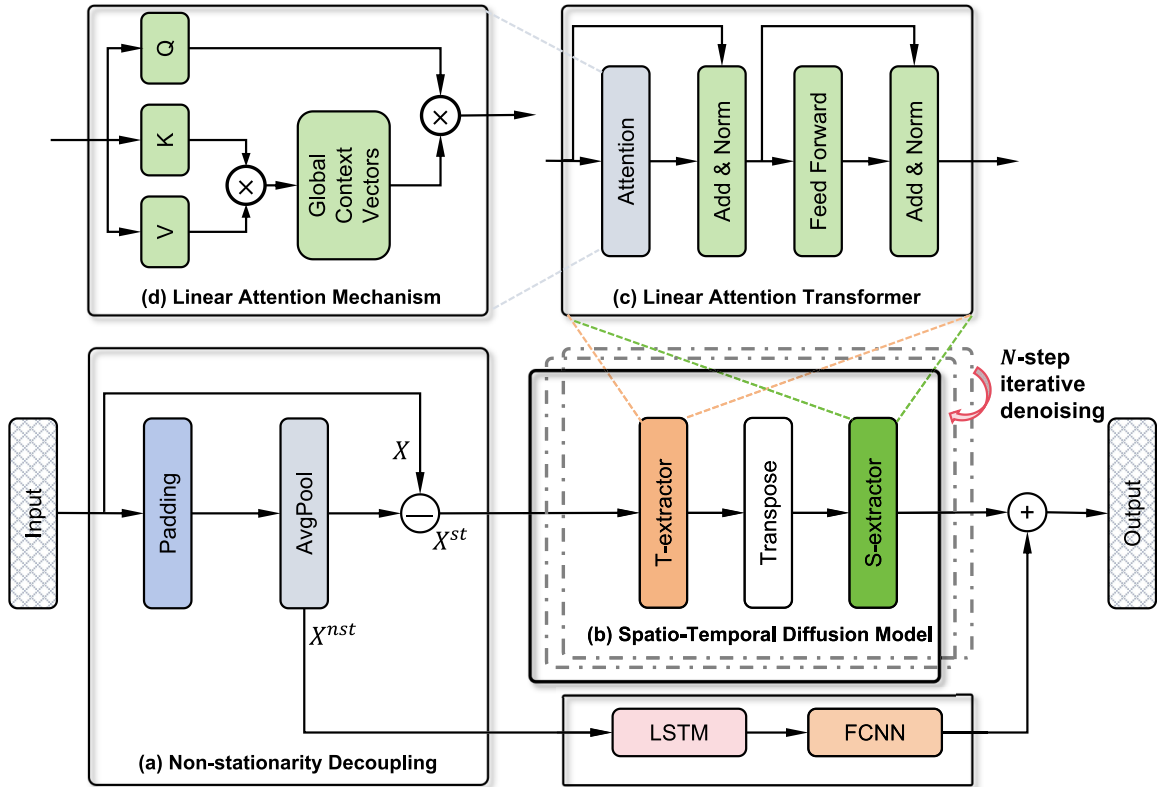


Fig. 3. Overview of the proposed DST-ND. (a) **Non-stationarity Decoupling** module extracts the non-stationary component from the original input data and separates the stationary component. (b) **Spatio-Temporal Diffusion Model** employs a diffusion model to perform reverse denoising on the stationary component, thereby generating the predicted stationary signal. Temporal and spatial dependencies are then extracted using the T-extractor and S-extractor, respectively. The detailed structure of STDM is illustrated in Fig. 5. (c) **Linear Attention Transformer** is used as both the T-extractor and S-extractor. When the input dimension corresponds to the temporal axis, LAT extracts temporal dependencies; otherwise, it captures spatial dependencies. (d) **Linear Attention Mechanism** is a linear-complexity attention mechanism.

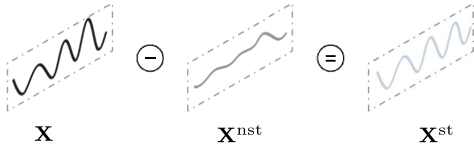


Fig. 4. Visualization of the original input $\mathbf{X}$, the non-stationary component $\mathbf{X}^{\text{nst}}$, and the stationary component $\mathbf{X}^{\text{st}}$.

D. Loss Function and Last Channel Prediction Result

The total loss sums the non-stationary and stationary losses and is optimized via back-propagation:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{nst}} + \mathcal{L}_{\text{st}}. \quad (26)$$

The channel prediction outputs are $\hat{\mathbf{Y}} = \hat{\mathbf{Y}}^{\text{nst}} + \hat{\mathbf{Y}}^{\text{st}}$, where $\hat{\mathbf{Y}} = \{\hat{\mathbf{x}}_{1:K}^{t+1}, \dots, \hat{\mathbf{x}}_{1:K}^{t+T}\} \in \mathbb{R}^{K \times T}$. Finally, we turn it into the original complex form $\hat{\mathbf{H}} = [\hat{\mathbf{H}}^{t+1}, \dots, \hat{\mathbf{H}}^{t+T}] \in \mathbb{C}^{T \times N_t \times N_r}$ of the channel according to the inverse process of (17).

V. SPATIO-TEMPORAL DIFFUSION MODEL

In this section, we introduce the key processes and modules of STDM.

A. Forward Diffusion Process & Training

For the stationary component $\mathbf{Y}^{\text{st}}$ obtained from NS-Dec, denote it as $\mathbf{x}_0$. STDM represents $\mathbf{x}_0 \sim q_{\mathcal{X}}(\mathbf{Y}^{\text{st}})$ in the training vector space $\mathcal{X} = \mathbb{R}^D$, and lets $p_{\theta}(\mathbf{x}_0)$ denote the

Algorithm 1 Training of STDM

Require: Distribution of training data $q(\mathbf{x}_0)$, conditional observations $\mathbf{x}_0^{\text{cond}}$ and covariant c .

Ensure: Trained denoising network ϵ_{θ} .

repeat

$n \sim \mathcal{U}(1, N)$, $\mathbf{x}_0 \sim q(\mathbf{x}_0)$, and $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

$\mathbf{x}_n = \sqrt{\alpha_n} \mathbf{x}_0 + \sqrt{1 - \alpha_n} \epsilon$

Take gradient step on $\nabla_{\theta} \|\epsilon - \epsilon_{\theta}(\mathbf{x}_n, n, \mathbf{x}_0^{\text{cond}}, c)\|_2^2$

until converged

probability density function (PDF). The goal of p_{θ} is to approximate $q_{\mathcal{X}}(\mathbf{Y}^{\text{st}})$. As shown in Algorithm 1, via a similar derivation in Section II-A, we have that the conditional variant of the objective (6) for noise index n is given by the following simplification of (9):

$$\mathcal{L} = \mathbb{E}_{n, \mathbf{x}_0, \mathbf{x}_0^{\text{cond}}, \epsilon} \left[\|\epsilon - \epsilon_{\theta}(\mathbf{x}_n, n, \mathbf{x}_0^{\text{cond}}, c)\|_2^2 \right], \quad (27)$$

where $\mathbf{x}_0^{\text{cond}}$ is the historical CSI used as a condition for the denoising network, and c denotes the covariant (in our method, c is the time frame information of the CSI).

B. Reverse Denoising Process & Sampling

Through the training of (27), a trained denoising network ϵ_{θ} will be obtained. Under ideal conditions, the distribution of the output of ϵ_{θ} for each diffusion step n will approximate the distribution of the added Gaussian noise ϵ .

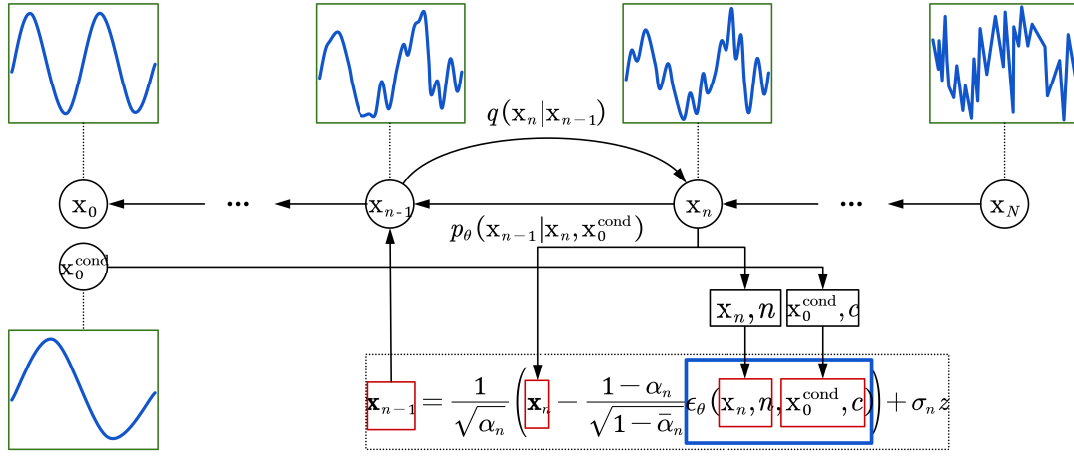


Fig. 5. Overview of the spatio-temporal diffusion model (STDM), with the structure of the denoising network ϵ_θ illustrated in Fig. 6.

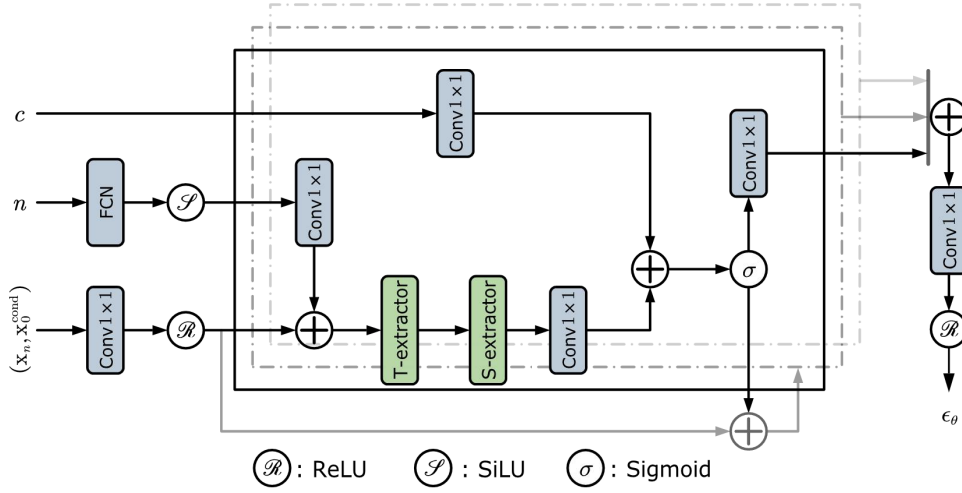


Fig. 6. Structure of the denoising function ϵ_θ in STDM. The T-extractor and S-extractor share an identical architecture, differing only in their input dimensions—one processes the time dimension, while the other focuses on the space dimension.

Algorithm 2 Sampling of STDM

Require: Trained denoising network ϵ_θ , conditional observations $\mathbf{x}_0^{\text{cond}}$ and covariant c .

Ensure: Predicted $\mathbf{x}_0$, i.e. the stationary component $\hat{\mathbf{Y}}^{\text{ST}}$.

```

 $\mathbf{x}_N \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
for  $n = N$  to 1 do
  if  $n = 1$  then
     $\mathbf{z} = \mathbf{0}$ 
  else
     $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
   $\mathbf{x}_{n-1} = \frac{1}{\sqrt{\alpha_n}} \left( \mathbf{x}_n - \frac{\beta_n}{\sqrt{1-\alpha_n}} \epsilon_\theta(\mathbf{x}_n, n, \mathbf{x}_0^{\text{cond}}, c) \right) + \sigma_n \mathbf{z}$ 

```

For sampling of STDM, as shown in Algorithm 2 and Fig. 5, the denoising process is carried out through a trained denoising network ϵ_θ , which iterates for N steps in the reverse direction using

$$\mathbf{x}_{n-1} = \frac{1}{\sqrt{\alpha_n}} \left(\mathbf{x}_n - \frac{\beta_n}{\sqrt{1-\alpha_n}} \epsilon_\theta(\mathbf{x}_n, n, \mathbf{x}_0^{\text{cond}}, c) \right) + \sigma_n \mathbf{z}, \quad (28)$$

where n ranges from N to 1, and $\mathbf{x}_0^{\text{cond}}$ is based on the past CSI as a condition. The process is similar to (11), gradually removing noise to obtain the final CSI. The denoising network ϵ_θ is the core of the diffusion model, and the next section will specifically introduce its implementation and how STDM extracts spatio-temporal correlations.

C. Spatio-Temporal Awareness Modeling

The denoising network ϵ_θ extracts the temporal and spatial correlations of the channel, as shown in Fig. 6. Unlike conventional methods that focus only on temporal coherence, STDM incorporates spatial correlation. This spatial correlation relates to the individual CSI matrix, considering the feature correlations among the K elements within the CSI matrix. Therefore, we decompose the mapping function (19) into f_{Θ_1} and f_{Θ_2} , which serve as the mapping functions for time and space, respectively. Our first step is akin to that depicted in (21) and is updated by

$$\hat{\mathbf{x}}[k]^{t+1}, \dots, \hat{\mathbf{x}}[k]^{t+T} = f_{\Theta_1}(\mathbf{x}[k]^{t-L+1}, \dots, \mathbf{x}[k]^t), \quad (29)$$

where k ranges from 1 to K , and $(\hat{\mathbf{x}}[1 : K]^{t+1}, \dots, \hat{\mathbf{x}}[1 : K]^{t+T}) \in \mathbb{R}^{K \times T}$. Then, for the output of the previous step, we transpose it by

$$\begin{aligned} &(\mathbf{x}[1]^{(t+1:t+T)}, \dots, \mathbf{x}[K]^{(t+1:t+T)}) \\ &= (\hat{\mathbf{x}}[1 : K]^{t+1}, \dots, \hat{\mathbf{x}}[1 : K]^{t+T})^\top \in \mathbb{R}^{T \times K}. \end{aligned} \quad (30)$$

Then, we model the spatial correlation by

$$\hat{\mathbf{x}}[1]^{t+i}, \dots, \hat{\mathbf{x}}[K]^{t+i} = f_{\Theta_2}(\mathbf{x}[1]^{t+i}, \dots, \mathbf{x}[K]^{t+i}), \quad (31)$$

where i ranges from 1 to T . To achieve the fitting, we ultimately need to transpose the output of f_2 to match the original input format.

Temporal/Spatial correlation extractor: LAT is used as the backbone of the Temporal correlation extractor (hereinafter abbreviated as T-extractor) and Spatial correlation extractor (hereinafter abbreviated as S-extractor) in (29) and (31), respectively, as

$$f_{\Theta_1} = \text{T-extractor}(L), \text{ and } f_{\Theta_2} = \text{T-extractor}(K), \quad (32)$$

where the inputs L (time dimension) and K (space dimension) correspond to the dimensions processed by LAT in the T-extractor and S-extractor, respectively. They are responsible for extracting temporal and spatial dependencies, respectively, as illustrated in Fig. 6.

VI. SIMULATION RESULTS

In this section, we present the performance comparison between the proposed method for channel prediction and existing methods.

A. Simulation Setup

In simulation, we consider the outdoor massive NLOS high-mobility scenario. Under NLOS conditions, the propagation path between the transmitter and receiver is obstructed by objects such as buildings and trees, preventing direct signal transmission. Instead, the signal reaches the receiver through diffraction, reflection, scattering, and other indirect means. The NLOS scenario typically encompasses a variety of signal measurements across different physical environments, including signal strength, arrival time, and CSI. The dataset consists of two subsets corresponding to speeds $v = 30$ km/h and $v = 60$ km/h, with 2 receiver antennas and 8 transmitter antennas; their specific details are presented in Table III.

Baselines: In this paper, we select four deep learning methods as the baseline models for comparison with DST-ND. They are Parallel Transformer [4], Rolling LSTM [15], STNN [39], Rolling RNN [12], LLM4CP [16] and STEM [17], all of which have demonstrated outstanding performance in the channel prediction task. Furthermore, to compare the performance with non-deep learning methods, we have included four non-deep learning methods. They are also labeled as Prony Vector (PVEC) [6], Auto-Regressive Integrated Moving Average (ARIMA) [40], Kalman Forecaster [41], and Fast Fourier Transform (FFT) [42]. These methods are training-free and use the rolling prediction approach to sequentially obtain the CSI for each future frame.

TABLE III
NLOS DATASET DESCRIPTIONS

Parameter	NLOS30	&	NLOS60
Speed	30 km/h	&	60 km/h
Channel type	UMa		
UE distribution	outdoor-NLOS		
Number of sectors	7		
Number of sectors per cell	3		
Number of UEs per sector	10		
Simulation drop	5		
Samples Per UE	153	&	308
Total samples	160650	&	323400
Duplex, Waveform	FDD, OFDM		
Multiple access	OFDMA		
Frequency Range	FR1, 2GHz		
Inter-BS distance	200m		
Channel model	According to TR 38.901		
BS Tx power	44dBm		
BS antenna height	25m		
UE antenna height & gain	Follow TR 36.873		
Numerology	Slot/non-slot: 14 OFDM symbol slot		
SCS	15kHz		
Simulation bandwidth	52 RB		
Subband bandwidth	4 RB		
Spatial consistency	Enable		
Sample interval	5ms		

TABLE IV
SIMULATION SETTINGS

Hyperparameter	Value
History frames L	25 (NLOS30) 40 (NLOS60)
Prediction frames T	{3,6,9,12,15} (NLOS30) {5,10,15,20,25} (NLOS60)
Training epochs	40
Learning rate	0.001
Number of sampling	75
Diffusion steps	100
Kernel size	5
Batch size	64
Optimizer	Adam (betas=(0.9, 0.999))

Metric: Performance is measured by Normalized Mean Square Error (NMSE), where a larger negative value indicates smaller prediction errors and superior model performance. The core advantage of the NMSE lies in eliminating the influence of units and scale, enabling a more objective assessment of the relative error of a prediction model. The calculation of NMSE follows the method described in [4], which is calculated by

$$\begin{aligned} \text{MSE} &= \sum_{t=1}^T \sum_{i=1}^{N_t} \sum_{j=1}^{N_r} \left\| h_{i,j}^{(t)} - \hat{h}_{i,j}^{(t)} \right\|^2, \\ \text{Power} &= \sum_{t=1}^T \sum_{i=1}^{N_t} \sum_{j=1}^{N_r} \left\| h_{i,j}^{(t)} \right\|^2, \\ \text{NMSE} &= \text{MSE}/\text{Power}, \end{aligned} \quad (33)$$

where $h_{i,j}^{(t)} \in \mathbb{C}$ denotes the value to be predicted in the target $\mathbf{H}^t \in \mathbb{C}^{N_t \times N_r}$, corresponding to the i -th transmitter antenna and the j -th receiver antenna, $\hat{h}_{i,j}^{(t)} \in \mathbb{C}$ represents the predicted result obtained by the neural network, and T denotes the prediction horizon.

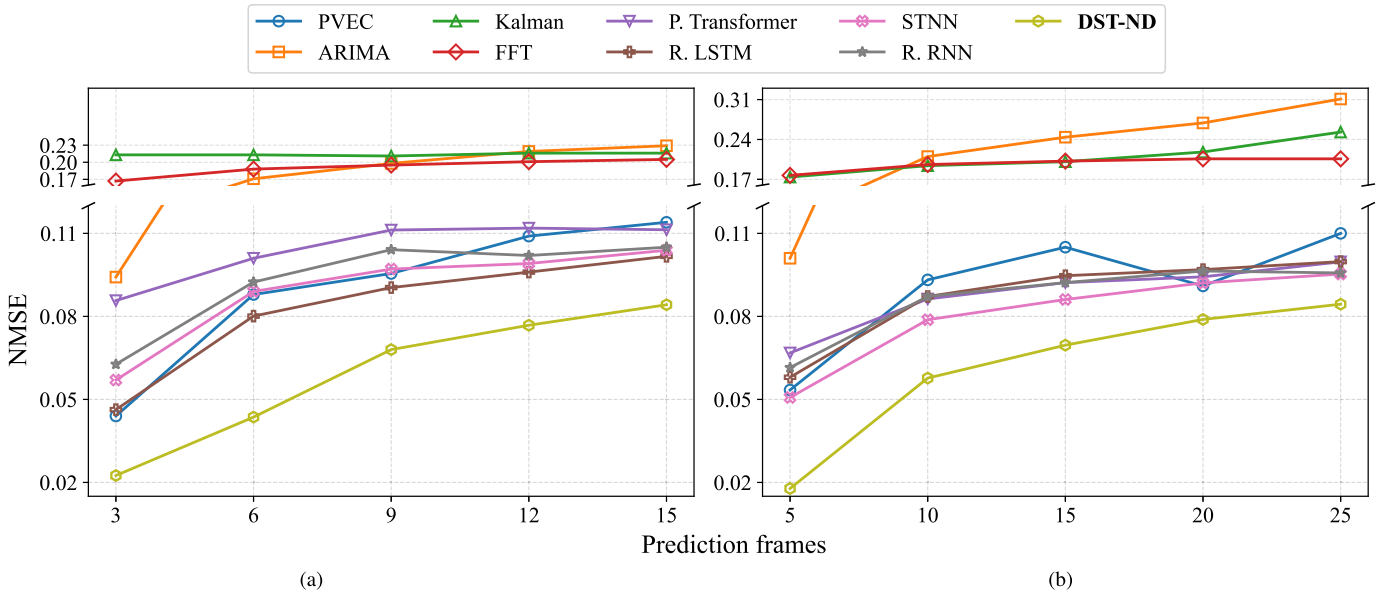


Fig. 7. Comparison of NMSE results between the proposed DST-ND and others baseline models.

TABLE V

IN THE THREE COMPONENTS OF ORIGINAL, NON-STATIONARY COMPONENT AND STATIONARY COMPONENT BEFORE AND AFTER DECOUPLING, THE PROPORTION OF CHANNELS REMAINING STATIONARITY

Dataset	NLOS30	NLOS60
Original	44.1%	38.7%
Non-stationary component	20.7%	11.7%
Stationary component	91.4%	99.2%

Environment and Simulation Setting: All experiments in this paper were implemented using PyTorch [43] with Python 3.12, trained using the Adam [44] optimizer, and executed on a single NVIDIA RTX A6000 GPU with 48 GB memory. Table IV shows the simulation setting.

B. Effectiveness of NS-Dec

Table V presents the stationarity proportions of channels before and after applying the proposed NS-Dec. In the original channel sequences, only 44.1% (NLOS30) and 38.7% (NLOS60) of the channels exhibit stationarity. After applying NS-Dec, the decoupling separates the signal into non-stationary and stationary components. The stationary component shows a significantly higher proportion of stationary channels, 91.4% and 99.2%, respectively, demonstrating that NS-Dec effectively isolates the stationary behavior of the channel. Conversely, the non-stationary component retains only 20.7% and 11.7% stationary channels, confirming that the dynamic variations have been concentrated into this component. These results highlight the strength of NS-Dec in disentangling mixed signal behaviors.

C. Results and Analysis

This section validates the effectiveness of the proposed DST-ND method and compares its performance with several other baselines.

TABLE VI

STATISTICAL SIGNIFICANCE TEST RESULTS. NMSE COMPARISON BETWEEN OUR METHOD AND THE LATEST BASELINES, LLM4CP AND STEM, ON THE NLOS30 AND NLOS60 DATASETS UNDER DIFFERENT PREDICTION HORIZONS

Dataset	Frames	Ours	LLM4CP	STEM
NLOS30	3	0.0225 ± 0.0011	0.0276	0.0387
	6	0.0436 ± 0.0018	0.0650	0.0680
	9	0.0680 ± 0.0016	0.0904	0.0865
	12	0.0768 ± 0.0008	0.0880	0.0932
	15	0.0842 ± 0.0021	0.0972	0.0963
NLOS60	5	0.0178 ± 0.0012	0.0393	0.0380
	10	0.0577 ± 0.0017	0.0700	0.0633
	15	0.0696 ± 0.0019	0.0795	0.0798
	20	0.0789 ± 0.0013	0.0885	0.0856
	25	0.0844 ± 0.0023	0.0894	0.0904

1) Performance Comparison on NLOS Datasets: Fig. 7 presents the NMSE comparison among prediction methods on NLOS30 and NLOS60 datasets. Non-deep learning methods (ARIMA, Kalman Forecaster, FFT) exhibit fundamental limitations: ARIMA's assumption of stationarity leads to high prediction variance in high-speed scenarios, while Kalman and FFT methods struggle with unknown noise distributions. In contrast, DST-ND achieves 67.6%-72.8% improvement over these methods, demonstrating the effectiveness of our decoupling mechanism. Compared to deep learning baselines, DST-ND delivers 13.6%-36.7% improvement across both datasets. This superior performance stems from two key innovations: (i) non-stationarity decoupling, which separates trend-drift from channel fluctuations and applies tailored modeling strategies to each component, and (ii) STDM's diffusion-based spatio-temporal correlation extraction, which effectively captures both spatial and temporal dependencies in the channel dynamics.

To validate the statistical significance of performance improvements, Table VI presents the results from three independent runs using different random seeds. The table reports both the mean NMSE values and standard deviations across

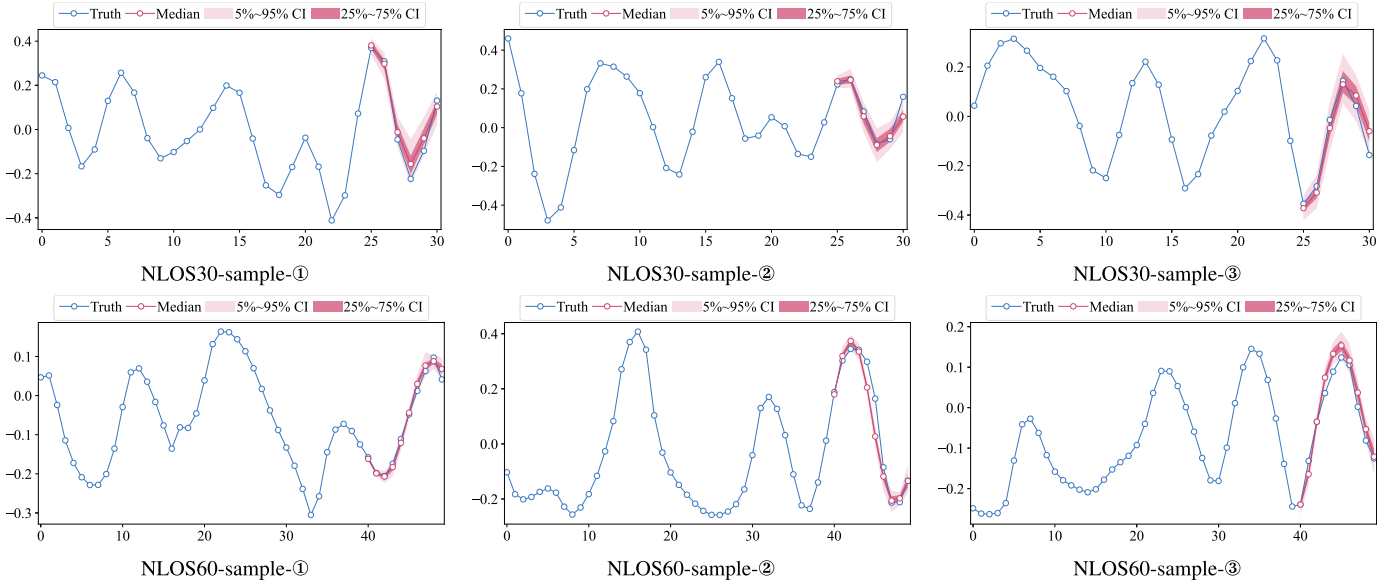


Fig. 8. Visualization samples of DST-ND. Top row shows three NLOS30 cases; bottom row shows three NLOS60 cases.

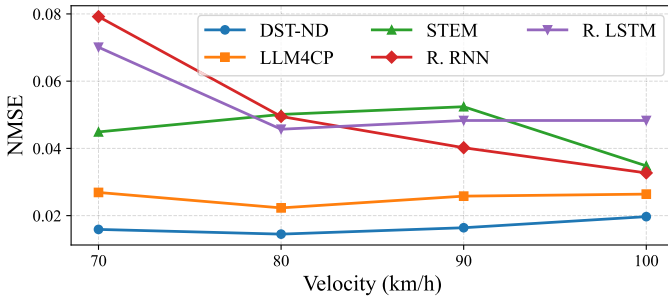


Fig. 9. NMSE comparison on the QuaDRiGa datasets.

these runs, demonstrating the robustness and consistency of DST-ND. The narrow standard deviations indicate that our method produces stable predictions across multiple random initializations, confirming the reliability of the proposed approach.

2) *Performance Comparison on QuaDRiGa Datasets:* We adopt the channel generator QuaDRiGa [45] to simulate time-varying CSI datasets that comply with 3GPP standards. The dataset covers multiple high-mobility scenarios with speeds of 70 km/h, 80 km/h, 90 km/h and 100 km/h. Fig. 9 reports the NMSE comparison of forecasting methods on these datasets. Our approach delivers outstanding prediction performance while simultaneously providing probabilistic forecasts. Compared with the NLOS dataset, QuaDRiGa features much higher mobility and more rapidly varying channels, making the prediction task considerably more challenging. Beyond prediction accuracy, our method additionally offers uncertainty estimation, which is essential for ensuring the reliability of practical communication systems.

3) *Probabilistic Prediction & Visualization Analysis:* Fig. 8 shows probabilistic forecasts on NLOS30 and NLOS60. The predicted 25% – 75% and 5% – 95% confidence intervals tightly cover the ground-truth CSI, indicating both accurate point predictions and calibrated uncertainty. Moreover, Fig. 11 visualizes three representative NLOS30 samples after

TABLE VII
ABLATIONS ON DST-ND, TOTAL 4 CASES

Ablation	NS-Dec	T-extractor	S-extractor	Exchange
①	×	✓	✓	×
②	✓	×	✓	×
③	✓	✓	×	×
④	✓	✓	✓	✓

our non-stationarity decoupling. Real-world non-stationarity is often coupled (abrupt changes, gradual drift, and multipath effects may co-exist). The three cases roughly correspond to: (i) abrupt changes, (ii) gradual drift, and (iii) multipath effects. Our moving average decoupling is a general approach applicable to various non-stationary scenarios, though it may be particularly well-suited for handling gradual drift, as the averaging operation naturally smooths low-frequency trend components while preserving their directionality. This separation supports the observed gains in downstream prediction.

D. Ablation Experiment

We summarize all ablations up front: ① remove NS-Dec (no decoupling, STDM handles all data); ② remove the T-extractor; ③ remove the S-extractor; ④ swap backbones so STDM models the non-stationary component while LSTM handles the stationary component; plus a backbone choice test for non-stationary modeling and a diffusion sampling-count study (efficiency vs. diversity). Table VII lists the first four settings.

1) *w/O. NS-Dec:* In the first ablation setting, we retain the spatio-temporal correlation modeling but remove the NS-Dec. Specifically, instead of decoupling the original CSI sequence into stationary and non-stationary components, we directly feed the raw CSI sequence into STDM for unified modeling and prediction. As shown by the green bars in Fig. 10, removing the NS-Dec leads to a performance degradation averaging

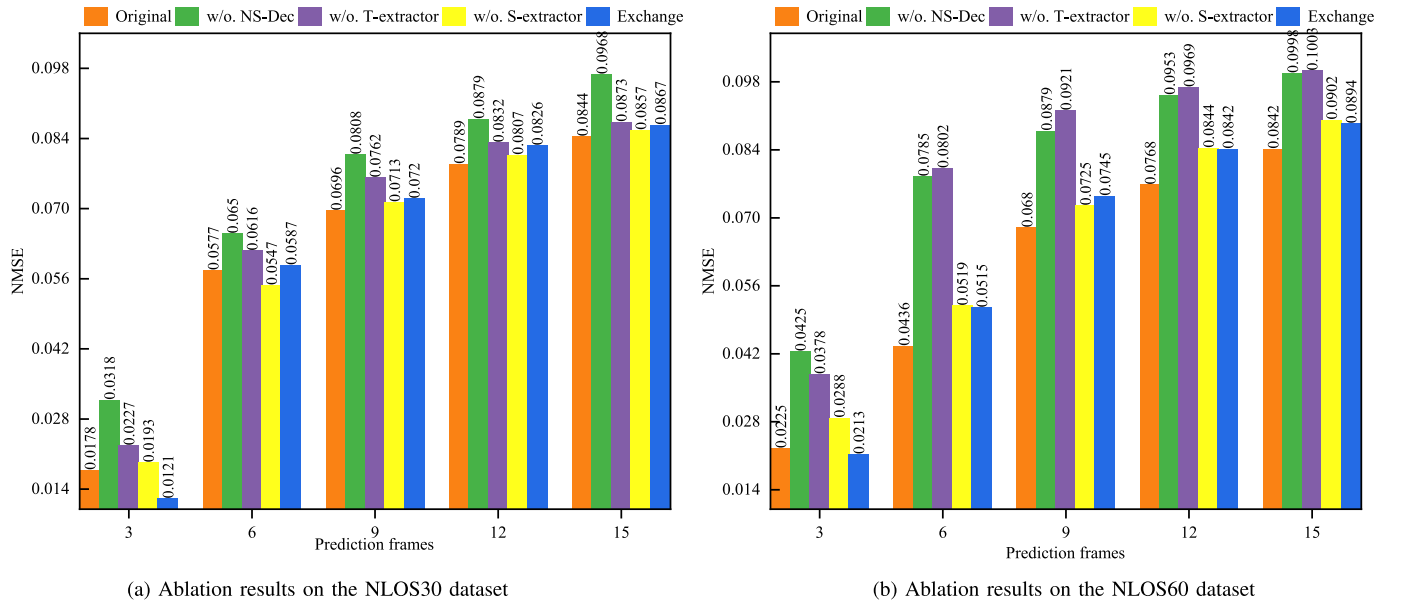


Fig. 10. Ablation experiments were conducted under identical experimental settings and included five prediction horizons.

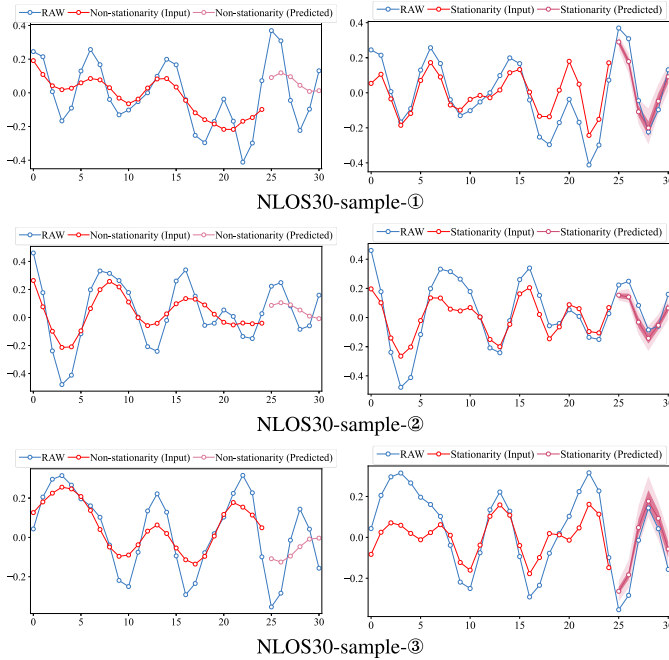


Fig. 11. Decoupling on NLOS30. Each row shows one sample, with two components placed side by side.

37.4% across different datasets and prediction horizons. This wide range of decline illustrates the critical role of decoupling in improving prediction accuracy. The decoupling process effectively separates long-term trends (non-stationary component) from short-term fluctuations (stationary component), allowing each to be modeled with specialized submodules. By isolating and explicitly handling the different statistical properties of channel components, the model can better capture the inherent structure of CSI dynamics. In contrast, applying a single modeling strategy to the entire CSI sequence—without decoupling—can lead to suboptimal learning, especially under conditions of strong temporal variation or mixed-frequency fading. Therefore, this experiment confirms that the proposed

NS-Dec is essential for enhancing the prediction capability of DST-ND, particularly in complex and fast-varying wireless environments.

2) *w/o. T-Extractor*: The second ablation case involves removing the time layer, thereby eliminating the extraction of temporal correlations. As shown by the Purple bars in Fig. 10, this modification leads to a significant degradation in performance. Specifically, the absence of temporal correlation modeling results in an average prediction accuracy drop of approximately 28.5%. This finding highlights the critical role of temporal dependencies in channel prediction. It also explains why nearly all existing methods preserve temporal modeling components. In sequential data such as CSI, the channel state at each frame is highly correlated with that of the previous frames, making temporal features one of the most essential elements for accurate forecasting.

3) *w/o. S-Extractor*: In the third ablation case, we remove the spatial modeling layer, retaining only the temporal correlation extraction. In this case, the model no longer captures spatial dependencies among antennas or subcarriers. As shown by the yellow bars in Fig. 10, this results in an average performance degradation of approximately 8.0%, with the worst-case drop reaching up to 28.0%. These results demonstrate the necessity of modeling spatial correlations in time-varying channels. Although temporal correlation remains the dominant factor in sequence prediction, spatial structures—such as correlations across antennas or subcarriers—carry valuable information that contributes to improved prediction accuracy. Moreover, in high-mobility scenarios, where channel aging becomes more severe due to rapid spatial variations, enhancing spatial correlation modeling is essential. Robust spatial feature extraction enables the model to better capture the dynamic structure of the wireless channel, thereby mitigating prediction errors caused by channel decorrelation. This highlights the critical role of spatiotemporal coupling in accurate CSI prediction under realistic mobility conditions.

TABLE VIII

BACKBONE NETWORK COMPARISON FOR NON-STATIONARY COMPONENT MODELING AND NMSE PERFORMANCE ON NLOS30 DATASET

Model	FCNN	RNN	LSTM	MLP	Transformer
NMSE	0.058	0.046	0.044	0.047	0.047

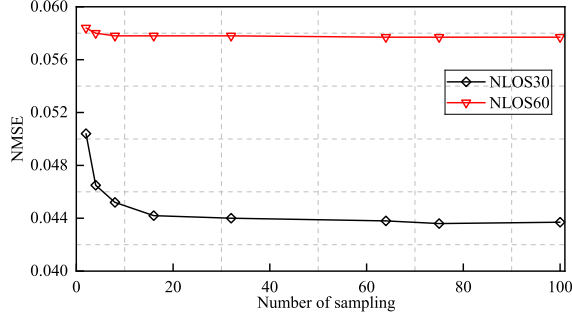


Fig. 12. NMSE results vary with different number of sampling.

4) *Exchange Non-Stationarity/Stationarity Modeling*: To investigate whether the STDM is more sensitive to stationary or non-stationary components, we conducted a modeling swap experiment, where the roles of STDM and the fully connected network were reversed. Specifically, we let STDM model the non-stationary component and used the LSTM to model the stationary component. Experimental results show that the swapped configuration achieves slightly better performance only when the prediction horizon is very short. In contrast, under most prediction settings, the original design consistently outperforms the swapped one. This suggests that STDM, which is based on denoising diffusion and spatiotemporal correlation extraction, is more effective when applied to stationary components of the channel. In contrast, the non-stationary component, which typically exhibits simpler trend patterns, can be adequately modeled by an LSTM.

5) *Backbone Network Selection for Non-Stationary Modeling*: Table VIII compares different backbone networks for non-stationary component modeling. LSTM achieves 0.044 NMSE, which is on par with RNN, MLP, and Transformer (0.046-0.047 NMSE). The comparable performance across these architectures indicates that LSTM provides sufficient modeling capability for the non-stationary component while maintaining computational efficiency.

E. Impact of Number of Sampling

We study the effect of diffusion sampling count. Fewer samples improve efficiency but reduce diversity; more samples tighten the distribution. Fig. 12 shows that on NLOS30, results converge after roughly 64 samples, while NLOS60 is largely insensitive to the count. The wider CI on NLOS30 versus NLOS60 (Section VI-C3) aligns with its shorter observation window ($L=25$ vs. $L=40$), making it more sample-hungry. Choose the sampling count per dataset to trade off diversity and cost.

Overall, decoupling and temporal extraction drive the largest gains (removing NS-Dec or the T-extractor causes the steepest

Algorithm 3 Inference Procedure of the Proposed Method

Require: Input past CSI observations $\mathcal{H} \in \mathbb{C}^{K \times L}$

Ensure: Predicted future CSI $\hat{\mathcal{H}} \in \mathbb{C}^{K \times T}$ with probabilistic forecasts

- 1: **// Step 1: Input preparation**
- 2: Vectorize $\mathcal{H}$ to feature representation $\mathbf{X} \in \mathbb{R}^{K \times L}$
- 3: **// Step 2: Non-stationarity decoupling**
- 4: Pad $\mathbf{X}$ symmetrically: $\mathbf{X}' = \text{Padding}(\mathbf{X})$
- 5: Extract trend via moving average: $\mathbf{X}^{\text{nst}} = \text{AvgPool}(\mathbf{X}')$
// Non-stationary component
- 6: Isolate fluctuations: $\mathbf{X}^{\text{st}} = \mathbf{X} - \mathbf{X}^{\text{nst}}$ // Stationary component
- 7: **// Step 3: Non-stationary component modeling with 2-layer LSTM**
- 8: Initialize LSTM hidden states: $\mathbf{s}_0, \mathbf{c}_0 \leftarrow \mathbf{0}$
- 9: **for** $\tau = 1$ to L **do**
- 10: $\mathbf{s}_\tau, \mathbf{c}_\tau = \text{LSTM}(\mathbf{x}_\tau^{\text{nst}}, \mathbf{s}_{\tau-1}, \mathbf{c}_{\tau-1})$ // Update hidden and cell states
- 11: Collect hidden states: $\mathbf{S} = [\mathbf{s}_1, \dots, \mathbf{s}_L] \in \mathbb{R}^{d_h \times L}$
- 12: Project to prediction horizon: $\hat{\mathbf{Y}}^{\text{nst}} = \mathbf{W}_p \mathbf{S} + \mathbf{b}_p$ // Linear projection to T -step forecast
- 13: **// Step 4: Stationary component modeling with diffusion**
- 14: Initialize Gaussian noise: $\mathbf{x}_N \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ // Random sample
- 15: Set conditioning information: $\mathbf{x}_0^{\text{cond}} = \mathbf{X}^{\text{st}}$ and c from timestamp // Encode context
- 16: **// Step 5: Reverse diffusion sampling**
- 17: **for** $n = N$ **down to** 1 **do**
- 18: **if** $n = 1$ **then**
- 19: $\mathbf{z} = \mathbf{0}$ // Deterministic final step
- 20: **else**
- 21: $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ // Stochastic steps for diversity
- 22: Predict noise: $\epsilon_\theta(\mathbf{x}_n, n, \mathbf{x}_0^{\text{cond}}, c)$ // Spatio-temporal extraction
- 23: Denoise: $\mathbf{x}_{n-1} = \frac{1}{\sqrt{\alpha_n}} \left(\mathbf{x}_n - \frac{\beta_n}{\sqrt{1-\alpha_n}} \epsilon_\theta \right) + \sigma_n \mathbf{z}$ // Iterative refinement
- 24: Extract denoised sample: $\hat{\mathbf{Y}}^{\text{st}} = \mathbf{x}_0$ // Final stationary prediction
- 25: **// Step 6: Combine predictions**
- 26: Reconstruct full prediction: $\hat{\mathbf{Y}} = \hat{\mathbf{Y}}^{\text{nst}} + \hat{\mathbf{Y}}^{\text{st}}$ // Merge trend and fluctuations
- 27: Devectorize to CSI format: $\hat{\mathcal{H}} = \text{Devectorize}(\hat{\mathbf{Y}})$
- 28: **return** $\hat{\mathcal{H}}$ // Predicted CSI with uncertainty

drops), spatial extraction still yields clear benefits, and the original module-task pairing (STDM on stationary, LSTM on non-stationary) outperforms the swap. LSTM suffices for non-stationary modeling versus heavier backbones, and diffusion sampling needs roughly 64 draws for NLOS30 while NLOS60 is insensitive.

F. Complexity Analysis

Table IX presents the computational complexity comparison among different channel prediction methods on the NLOS30 dataset. Our method has 2.2M parameters, comparable to Rolling LSTM, and significantly fewer than LLM4CP (46.8M)

TABLE IX

COMPUTATIONAL COMPLEXITY COMPARISON ON NLOS30 DATASET WITH INPUT LENGTH 25 AND PREDICTION HORIZON 6. TRAINING TIME IS MEASURED PER EPOCH; INFERENCE TIME IS PER BATCH

Model	Parameters	Training (s)	Inference (s)
R. RNN	543.1K	13.449	0.013
R. LSTM	2.2 M	24.189	0.015
P. Transformer	6.2 M	6.385	0.008
LLM4CP	46.8 M	8.306	0.008
STEM	3.0 M	6.962	0.007
Ours	2.2 M	20.246	$N \times 0.013$

and P. Transformer (6.2M). The training time of DST-ND is 20.246 seconds per epoch, which is moderate among all methods. However, due to the iterative sampling process inherent in diffusion models, the inference time is $N \times 0.013$ seconds, where N denotes the number of diffusion sampling steps. This iterative nature introduces additional computational overhead during inference compared to single-pass prediction methods like STEM and P. Transformer. Despite the increased inference cost, DST-ND achieves superior prediction accuracy and provides probabilistic forecasts, which are essential for reliable communication system design. In practical deployment, the number of sampling steps N can be adjusted to balance prediction quality and computational efficiency.

VII. CONCLUSION

This paper proposes DST-ND, a novel diffusion-based approach for time-varying channel prediction that addresses the challenge of temporal distribution shift in high-mobility wireless environments through non-stationarity decoupling. The key innovation lies in the synergistic combination of two core components: a non-stationarity decoupling module that separates stationary and non-stationary components, and a spatio-temporal diffusion model that captures both spatial and temporal correlations within the denoising process. By tailoring the prediction strategy to each component's statistical characteristics, applying diffusion-based modeling to stationary components and LSTM-based modeling to non-stationary components, DST-ND achieves superior accuracy while providing probabilistic forecasts with uncertainty quantification. Experimental validation on multiple datasets demonstrates consistent improvements over existing methods, with performance gains ranging from 13.6% to 72.8% depending on the scenario. Ablation studies confirm the critical contribution of each module, demonstrating that the proposed design is well-justified and effective.

Future Work: We identify three promising directions. First, incorporate inter-subcarrier correlations and extend the framework to multi-user systems. Second, since diffusion denoising requires multiple iterative steps, we will explore fast sampling strategies such as DDIM [46] to reduce the number of diffusion steps, thereby lowering inference latency while maintaining prediction quality. Third, extract richer correlation features from time-varying channels to further improve prediction accuracy.

APPENDIX

DETAILED INFERENCE PROCEDURE OF DST-ND

Algorithm 3 presents the complete inference procedure of our proposed method. The algorithm consists of six main stages: (i) vectorization of the input CSI, (ii) non-stationarity decoupling to separate non-stationary and stationary components, (iii) LSTM-based modeling of the non-stationary component, (iv) initialization of the diffusion process with conditioning information, (v) iterative reverse diffusion sampling for the stationary component, and (vi) final reconstruction by combining both components. The diffusion sampling loop performs N iterations to progressively denoise the random sample; earlier steps introduce stochasticity for diversity, and the final step is deterministic to stabilize convergence. The conditioning mechanism ensures that the diffusion model respects the observed context, enabling reliable probabilistic forecasting.

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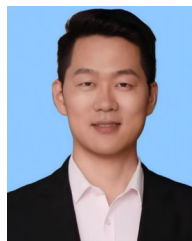
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